

Q. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Here are three common ways **Instagram** and **Zomato** may use **firewalls** and **packet filtering** to protect their servers and users.

1. Blocking Unauthorized Access

Firewalls allow only trusted network traffic to reach the servers. If someone tries to connect using unauthorized ports or suspicious IP addresses, the firewall blocks the request before it reaches the application.

2. Preventing DDoS (Distributed Denial-of-Service) Attacks

Both Instagram and Zomato can use firewalls and packet filtering to detect unusually high volumes of traffic from suspicious sources. Malicious traffic is filtered or rate-limited, helping keep the website and mobile app available for legitimate users.

3. Filtering Malicious Traffic

Packet filtering checks incoming and outgoing network packets based on rules such as IP address, port number, and protocol. Traffic that appears to be malicious—such as scanning attempts, fake requests, or packets from blacklisted IP addresses—is blocked to reduce the risk of attacks and misuse.